Housing First or Treatment First?

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Response to Referee 2

Conceptual and detail concerns, responses in red

Thank you for the report, and in particular for the two welfare margins in your second paragraph. One of them has become a feature of the revised model and the other has become an intergenerational linkage that the Associate Editor independently asked for. Both changes came from this report.

One note before the comments. The paper has a new title: it was submitted as *Housing First or Treatment First?* and is now *Housing First, Children Later*. The reason is the substance of the revision rather than a relabelling, and it will be clearest at C3, since the mechanism the title names is the one you asked for there.

Conceptual concerns

C1 The welfare results rest on the elasticity of drug use to housing access

> *"... all of the welfare results rest on whether the model has an accurate elasticity of drug use to housing access. The only empirical drug use elasticity mentioned is a price elasticity of drug use participation, but it isn't clear to me that this provides enough or the correct discipline for these policy experiments. I would like to see that the model matches some evidence on the response of drug use to housing access (see Milby, et al (2005) for cocaine users, but there are many papers on this topic.)"*

This is the crux, Referee 3 and the Associate Editor agree, and it is now a calibration target rather than an untested implication.

Milby et al. (2005) is the right trial for it, and it turns out to identify two things rather than one, which resolved a separate problem in the model.

- The *contrast between arms* identifies the response of abstinence to housing access, which is your requested elasticity.
- The *level in the no housing arm*, 4.48 consecutive weeks abstinent out of 24, or 18.7 per cent among homeless adults with a substance use disorder, identifies how costly abstention is for a household that already has the disorder. That parameter was previously asserted to be small enough to keep the drug choice interior, which is a statement about numerical convenience rather than about data. It also left the housing access moment undefined: if abstention is impossible for a household with a disorder, the no housing arm abstains zero per cent of the time and the ratio between arms is zero over zero.

One tension in this literature is worth reporting, because it looks like a contradiction and is not. Milby finds housing raises abstinence. The At Home / Chez Soi trial finds unconditional housing does nothing to substance use problem severity, 1.04 with a confidence interval of 0.89 to 1.21. Milby's housing was *contingent on abstinence*; At Home / Chez Soi's was *unconditional*. Contingency is doing the work, not housing. That distinction is exactly the paper's subject, and the two trials together are better evidence for it than either alone. This is now in the text.

[Built since the first draft of this letter.] It is now in the code and not only in the written model, and it sits in a different place from the one I first proposed. My first draft made the Stone-Geary shifter fall with homelessness, so that housing lowered use through a taste. I have retired that, for two reasons I would rather state than have you find. Housing in this model is a *choice*, so a preference that depends on it is a preference over a choice: a household would pick its dwelling partly to move its own taste, and a user would acquire a reason to choose the street because the drug is worth more there, which is close to the reverse of the selection the paper is about. And the sign is odd in any case, since the marginal value of the drug goes as $1/(b+v)$, so the channel would have worked by asserting that homelessness improves the experience of using.

The elasticity you asked for does not need a taste, and it is better identified without one. It now runs entirely through the *recovery hazard*: recovery from a substance use disorder is six times harder on the street than in a home. That is a transition probability rather than a preference, it is what a trial measures, and it is what your own citation reports. Milby et al. record abstinence by arm; abstinence by arm is a hazard. So the requested moment is targeted, the sign is right, and the flow utility is now the same function of consumption, housing and drugs in every state and after every choice, which is also what your comment C5 and the Associate Editor's A8 asked for. The two requests turn out to be the same request, and I had been answering them with different machinery.

C2 A large homeless population is a negative externality

> *"... a large homeless population, especially if many are addicted to drugs, creates a negative externality beyond the fiscal burden of providing services. ... One structural rental housing paper that considers the negative externality of homelessness (though doesn't have drug addiction) is Corbae, et al (2024). ... the externality seems very easy to introduce into the current model (though would require a new calibration moment)."*

Adopted, and it does more work than you may have anticipated, because it lets me delete something that three of the four reports objected to.

A disutility from the size of the homeless population now enters period utility. This replaces the loss of consortium adjustment entirely. Referee 1's comments 3(a) and 3(b) said that adjustment was applied after the model was solved, was therefore internally inconsistent, and rested on a parameter whose conservative value is zero. The Associate Editor's comment A9 said the same. Your externality answers all of those objections at once, because it sits *inside* the value function where households internalise it.

It also gives Referee 3's comment M2c a concrete answer. He asks where the market failure comes from if drug use is rational and forward looking. There are now two answers in the model: the fiscal externality, which was always there, and this one, which is not fiscal and which no individual household internalises when it chooses whether to use.

On Corbae, Glover and Nattinger: I cite them, and I am explicit about what does and does not transfer. Their externality is about neighbourhood housing *quality* and is identified off rent decontrol, so their functional form is not the right one to import. What transfers is the identification logic. I use a contingent valuation approach for the calibration moment instead.

[Still open.] It is in the model as written and *not yet in the code*, which I would rather say than imply otherwise: the term appears in the household's problem in the paper, and the Fortran does not yet carry it. Because the aggregate headcount is taken as given by every household, its absence cannot move a single policy function, so nothing in the calibration depends on it; what

it changes is the welfare comparison across steady states, which is where I use it, and that is why it has to be built before I report that comparison.

It is priced from the evidence you would want it priced from rather than asserted. Loubière et al. (2020) ask adults in eight European countries what they would pay in annual taxes to end homelessness through Housing First: €23 a year on average, which is 0.08 per cent of average annual earnings, and €57 among the 42 per cent willing to pay anything, 0.20 per cent. I take the *higher* figure, 0.20 per cent, as the baseline. Setting it high should favour Housing First, which houses more people, so a ranking that still favours Treatment First survives the assumption least kind to it; the lower figure is the other end of the range.

One choice inside it is worth flagging, because your comment mentioned addiction explicitly. The disamenity is on the homelessness *headcount alone*, with no extra weight on the homeless who carry a disorder. The survey asks what people would pay to end homelessness, so homelessness is what the number prices; loading addiction into the same term would price something the evidence was not about, and would also make the term move with a margin the paper is separately trying to measure.

The honest expectation is that it will not decide anything on its own: 0.08 per cent of earnings is small next to the fiscal and mortality channels. What it does is add to whichever policy houses more people, so it can only strengthen a housing result, never manufacture one, and reporting it across the range matters more than the point estimate.

C3 Parents suffer anxiety about their children

> *"... parents suffer anxiety that their children may become drug addicts and/or homeless in adulthood. ... the dynastic element seems like more of a stretch, but could be discussed (I think it would only amplify the welfare gains from reducing addiction)."*

I have done more than discuss it, because the Associate Editor's comment A6 pushes in the same direction from a different angle: he objects that the previous draft discussed intergenerational transmission in section 4.1.3 without any intergenerational linkage in the model, which made the discussion decorative. Your comment supplies exactly the linkage that would make it load bearing.

[**New result.**] This comment is the origin of the paper's central mechanism, and I would rather say so plainly than let it look like my own idea arriving late. You asked for it, the Associate Editor asked for the same thing from the other direction in A6, and between the two comments the channel and the defect were both identified. It is now what the paper is about, and it is why the paper carries a different title. A household that acquires the disorder, or that ends up on the street, passes a worse productivity draw to the child who replaces it, and counts only a fraction ω of that loss. The remainder is an externality, and it turns out to be the one that decides the policy comparison: at the same generosity and the same cost, Housing First weakly dominates Treatment First for every household taken one at a time, and the ranking reverses once the children of the households it fails to deter are counted. Each link in the chain is measured rather than assumed, and the transmission penalty is calibrated inside the model to reproduce the ratio of disorder rates between the children of parents with and without a disorder.

Two things about it are worth flagging, since you called the dynastic element a stretch.

It does not require tracking dynasties or a new state. A household's continuation splits into its own next period, weighted by survival, and the value of the child who replaces it, weighted by

the chance of dying and by ω. The child term is a *difference*, the value of a child born to this household against a child born to a clean and housed one, so it is zero in the reference state and negative otherwise, and what enters the decision is only the loss the parent's own state imposes.

And ω is not free. Every household in the model has a child, so there is no parent against non parent comparison to run in the benchmark; the moment is a counterfactual instead. In NSDUH 2024, past month use among parents is 0.845 of use among non parents, reweighted to the non parents' joint distribution of age, income, sex and marital status, and ω is set so that removing the child from the household's problem raises use by that same 18.3 per cent. The externality is therefore not assumed to be large. Its size is read off what parents already do.

Your prediction was that the dynastic element would amplify the welfare gains from reducing addiction rather than reverse anything. In this model it does more than amplify: it is what overturns a ranking that is otherwise unambiguous. I report the results at ω set to zero as well, so a reader can see exactly how much rests on it.

C4 The rich model is a black box; show the policy functions

> *"... it can be a bit of a black box. It would be helpful to have a section where the mechanisms of the model are explored more thoroughly. For example, it would be useful to show and explain intuitively a plot of the drug use policy function for addicts and non-addicts as their income and wealth vary."*

Done, as the figure you describe, and it is now part of a broader mechanism section that also serves Referee 1's comment 1(d) and Referee 3's comment M2b.

The section has three parts: the policy function figure you ask for; a channel shutdown ladder that re solves the model with one channel disabled at a time and reports what happens to the income gradient of use; and a decomposition of the sources of the externality. The three together are the paper's answer to the charge that the model is opaque, and they replace the static appendix that Referee 1 found was claiming more than it established.

C5 The utility specification bakes in an income contingent gain or loss

> *"It would also help to further explain the importance of the period utility specification in (13) which, I think, implies that drug addiction is more attractive for low income people ... it also bakes in an income-contingent gain/loss from becoming addicted by shifting the utility weight towards drug consumption away from non-drug consumption and housing."*

[Conceded.] You and Referee 3, in his comment M2a, identify the same problem, and it is a serious one: the previous specification made reducing substance use disorder and expanding housing mechanical welfare complements even with no incentive effect at all. Since the complementarity between generosity and conditionality is the paper's headline result, the specification was doing the work that the economics was supposed to do.

Preferences are now *fixed* across substance use disorder status: one set of weights, one subsistence parameter. Nothing about the utility function changes when a household acquires the disorder.

The Associate Editor suggested a milder version, keeping the weights invariant but letting the subsistence parameter vary by status. I tried that and it does not reproduce the target the specification was carrying, namely the gap between 44 per cent of income spent on drugs among people with a disorder and 11 per cent among experimenters: the subsistence parameter shifts the level of drug spending, but the ratio is governed by the curvature of the subsistence term, and opening a fourfold gap by moving it drives experimenters to a corner.

The reason preferences are fixed rather than merely fixed enough. There is an argument here that goes beyond fitting a moment, and it is what settled the specification. Any formulation that makes a household *more* susceptible to drugs by moving its preferences also makes that household *better off*, mechanically, because it is now getting more utility out of something it was going to consume anyway. Flow utility is Cobb-Douglas in $b + v$, so the marginal value of the drug is proportional to $1/(b + v)$: a smaller subsistence parameter raises the marginal value of using, raises desired intake, and raises the level of utility along with it. That sign is the wrong way round against everything known about substance use disorders, which are not experienced as a windfall by the people who have them, and it puts the welfare accounting on the wrong footing, because the policy that prevents a disorder is then taking something valuable away.

So the model varies *income* rather than utility. What the disorder changes is the budget and the hazards a household faces: lost earnings, lost employment, higher mortality, harder recovery. Every one of those is measured, and every one of them makes the household worse off, which is the direction the evidence points. The same argument governs what a child inherits, and it is worth saying because it is the paper's central mechanism: the child of a user inherits a worse productivity draw, not a stronger taste. Had it inherited the taste, its lifetime value would have gone *up*, and the specification would be asserting that a child of a parent with a disorder is better off for it.

With fully fixed preferences the 44 against 11 gap has to emerge endogenously, and it does, from forward looking restraint. A household without a disorder limits its use because use raises the probability of acquiring one. A household that already has one faces no such margin. The spending gap becomes a prediction rather than an assumption, and the welfare complementarity becomes an incentive effect rather than an artefact of the utility function. This is a stronger result than the one it replaces and I am grateful for the push.

[**Built since the first draft of this letter.**] Done in the code, where it previously was not, and taken one step further than you asked. The switch that raised the weight on drugs from 0.13 to 0.45 for a household with the disorder is off, so the weights on consumption, housing and drugs are the same function clean and addicted. I then found that the Stone-Geary shifter raised the same objection one step along, and it is now constant too: flow utility is the same function of consumption, housing and drugs in every state and after every choice. What the disorder costs runs entirely through mortality, employment, the recovery hazard and the budget, none of which is a preference.

C6 Optimal voucher design, and χ_H looks high

> *"The parameter χ_H governs the fiscal cost of having somebody unhoused and appears pretty high at 25% of average income. Would there be significant further welfare gains from broadening the voucher program by making higher incomes eligible or covering more of the rent (with the drug-free contingency)? My (perhaps wrong) reading is that the cost of having somebody unhoused is significantly higher than the rent on the smallest housing unit."*

Your reading is correct, it is not wrong, and it is the mechanism behind the near self financing result. The paper should be transparent about that rather than let it emerge as a surprise, so it is now stated in the text where the result is presented, and the sensitivity is in the body rather than an appendix.

On your first question, whether broadening the programme would deliver further gains, the paper now solves for the optimal voucher design rather than comparing a fixed handful of programmes,

varying the eligibility cutoff and the share of rent covered. Referee 3's minor comment 3 asks for the same thing from the rationing side, that the probability of receipt be varied continuously rather than jumped from rationing to universal cover, and the two are the same exercise.

The optimum is also where the channel decomposition is reported, because that is where it is most informative. The same policy is re solved with one channel switched off at a time: the child externality, the aggregate disamenity you asked for in C2, the labour supply response that carries the tax distortion, the drug price response, and the cost of administering the sobriety test. What that says is not only which policy wins but how much of the optimal design each channel is responsible for. The rungs do not commute, since switching the externality off changes who is poor and therefore how far the tax base moves, so a fixed order is reported together with the interaction term rather than a residual buried in whichever channel goes last.

On the level of χ_H, the Associate Editor's comment A15 points to Abramson and Van Nieuwerburgh as a comparison. Having read their paper, there is a point of fact worth recording: their 2,500 dollars per household per month for the literally homeless, and half that for the doubled up, is an *assumption* inside their calibration, stated as such with components listed in their footnote 11, rather than an independent estimate. So it cannot validate my number. I report the comparison, because a reader will want it, described accurately as a comparison of two calibration choices. Blended across their two groups it comes to about 1,775 dollars per month.

On optimal design, I now run the continuous sweep that Referee 3's minor comment 3 also asks for: the probability of voucher receipt is varied from the benchmark to universal, and welfare, homelessness, use and the labour tax are plotted along it. This shows whether universal provision is optimal or merely a corner, which is his question, and it shows where the gains from broadening eligibility run out, which is yours. I run the sweep separately with and without the sobriety condition, so the cross partial that the paper's headline rests on is visible as the gap between two curves rather than as a comparison of four table cells.

Detail concerns

D1 The housing grid

> *"... the housing grid. It has seven values but only three are calibrated ... The full grid is then listed in footnote 4 rather than a calibration table. My assumption before seeing the grid was that the bounds were calibrated and then the sizes were distributed uniformly ... But they are not uniformly spread out, at least in levels or ratios. So, how are the housing levels being chosen and does it matter for results? Could they just be spread uniformly without affecting the calibration and policy conclusions? If not then I think they need to be included as parameters in the calibration."*

The honest answer to the original question is that the grid was not chosen by a principle I could defend, which is why the footnote existed. The revision removes the problem rather than justifying it.

With homeownership cut at the Associate Editor's request in his comment A4, the housing state is homeless plus three rental sizes. Every point of that grid is now pinned by a moment, and the moments are the size distribution of occupied units from the American Community Survey, tenure by bedrooms: 13.21, 24.12 and 62.67 per cent across zero to one, two, and three or more bedrooms. All households rather than renters only, because the model has no homeownership

and housing is a pure flow, so an owner's analogue is imputed rent and the distribution over all occupied units is the right object. The middle category is the residual and is not targeted.

The grid now appears in the calibration table with its moments, not in a footnote. As a by product, the two housing cost burden moments that previously identified the grid are freed to serve as out of sample validation, which is a stronger thing to show than a moment the grid was fitted to.